

Oracle Externality in Prediction Markets Creates Adversarial Incentives Against Public Infrastructure

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Prediction markets on real world events resolve based on data produced by external oracle systems, frequently from public infrastructure maintained under non-adversarial threat models. The financial incentives these markets create for specific outcomes transform such infrastructure into attack surfaces whose hardening costs fall on operators and users outside the market. We describe the April 2026 manipulation of Météo-France sensor data for Polymarket weather contracts as an illustrative case of this pattern, situate it within existing discussions of prediction market harms, and argue that the mechanism represents a distinct class of externality not yet addressed by either the mainstream prediction market literature or the cryptocurrency-focused oracle security literature. We sketch implications for regulatory design and identify directions for further analysis. The paper articulates the mechanism rather than its formal resolution, as we treat this as an opening argument for a problem that we argue deserves dedicated attention.

Keywords: prediction markets, externalities, oracle manipulation, public infrastructure, algorithmic governance, regulation

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1 Introduction

In early April 2026, anomalous temperature readings from a Météo-France weather station on the perimeter of Paris Charles de Gaulle airport coincided with unusually profitable bets on the Polymarket prediction market [19]. On April 6th, the station's recorded maximum temperature spiked above 21°C in a pattern inconsistent with surrounding weather, shifting a Polymarket contract's resolution probability from approximately 5% to 95% in under an hour. The account placing the winning bet had been created two days earlier and netted approximately \$14,000 on an initial stake of a few tens of dollars. Nine days later, on 15 April, a similar spike to 22°C produced a further payout of over \$20,000. Météo-France subsequently filed a formal complaint, and Polymarket changed the data source for its Paris weather contracts.

The immediate facts of the incident are, frankly, absurd enough to obscure a structural observation. The manipulation did not require sophisticated attacks against cryptographic oracles or coordinated action across multiple trading accounts. It required a heat source (purportedly a blow dryer), knowledge of a sensor's location,

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and a willingness to drive to an airport. The attack was crude, and it succeeded twice before pattern-matching by amateur meteorologists on the *Infoclimat* forum triggered investigation [11]. The question is not how sophisticated attackers might exploit prediction market oracles, but rather how crude successful attackers already are.

We argue that the Météo-France incident demonstrates a novel class of externality created by prediction markets that has not been analytically articulated in the existing literature. Prediction markets on real-world events require oracles to resolve, and the oracles frequently depend on public infrastructure such as weather stations, election reporting systems, or public health databases. This infrastructure was constructed and maintained under threat models that assumed non-adversarial use. When prediction markets attach financial incentives to specific values of the data this infrastructure produces, they transform it into an attack surface. The costs of adversarial hardening, or of degradation if hardening does not occur, fall on infrastructure operators and on the broader population of infrastructure users, not on the market operator or market participants. We can call this the oracle externality.

Existing discussions of prediction market harms have noted specific aspects of this pattern without developing it as a distinct mechanism. Packin and Rabinovitz [15] argue that commercial prediction markets pose risks to democratic integrity, public health, and consumer welfare, and propose that platforms be required to internalize their externalities. Their analysis is a policy commentary rather than an analytical treatment as they identify the problem class without characterizing its structure. The cryptocurrency-focused oracle security literature [3] analyzes manipulation of on-chain governance oracles and price feeds, but its threat models presume digital attack surfaces and do not engage with physical infrastructure attacks. The mainstream prediction market literature [9, 21] focuses on information aggregation properties and generally treats externalities as positive rather than negative.

Our contribution is to articulate the oracle externality as a structural feature of prediction markets on real world events, distinguish it from adjacent phenomena (insider trading, market manipulation, digital oracle attacks), and argue for its policy relevance. We are explicit that this is a framing contribution. Formal modeling of equilibrium behavior and empirical estimation of externality magnitudes are directions for subsequent work, and here we argue that these directions are worth pursuing, which requires first establishing that the mechanism they address is novel.

2 The Mechanism

Consider a prediction market offering contracts that resolve based on data produced by some external infrastructure. For concreteness, take a contract that pays out if the maximum temperature at a specified weather station exceeds a specified threshold on a specified date. Market participants place bets with expected values depending on the eventual reading. The infrastructure operator (here, Météo-France) produces the reading through its normal operational procedures, which were designed to measure temperature accurately under assumptions about the physical environment and the behavior of parties with physical access to the sensor.

These assumptions included, historically, that no party had a financial incentive to cause the sensor to report a specific value. Temperature data users (such as meteorologists, aviators, agricultural planners, climate researchers) benefited from accurate measurement and had no incentive to distort it. Physical security of the sensor was therefore minimal, limited to protection against accidental damage, vandalism whose motivations were not data specific, and natural disturbances. The infrastructure was adequate for its threat model.

Prediction markets change the threat model, as a contract that pays out when the temperature exceeds a specific value creates, for the bettor holding that contract, a financial incentive to cause the temperature reading to exceed that value. The incentive exists regardless of whether the bettor intends to act on it, and it scales with the bet size

and the contract's payout structure. Across a population of market participants, some fraction will have both the capability and the disposition to act on the incentive. The attack in the Météo-France case required physical access to a sensor and the application of heat, and neither barrier was high.

The infrastructure operator bears the consequences. If the sensor is not hardened against manipulation, the infrastructure's outputs are degraded, with costs falling on all downstream users. If the sensor is hardened, the operator bears the cost of hardening, which was not part of the cost structure under which the infrastructure was originally provisioned. Either way, a cost is imposed on the infrastructure operator that did not exist before the prediction market created the attack incentive.

Several features of this structure distinguish it from adjacent phenomena. It is not insider trading as no party uses non-public information, and the attack succeeds against public information. It is not market manipulation in the conventional sense since the manipulation targets the oracle rather than the market's trading dynamics, and standard market manipulation enforcement mechanisms do not apply to attacks carried out physically on external infrastructure. It is not the cryptocurrency oracle attack problem analyzed in the DeFi security literature because the threat surface is physical rather than digital, and the defensive mechanisms developed for on-chain oracles face additional challenges when applied to physical measurement systems.

The closest structural analog in the existing literature is the manipulation of judgment-based financial benchmarks such as LIBOR, where reporting systems designed under assumptions of honest participation became attack surfaces once large derivative positions created incentives for distortion [7]. The oracle externality differs in that the benchmark operator and the market operator are entirely distinct entities, so that the manipulation incentive is created by one party and the costs are borne by another with no contractual relationship between them.

The most important feature, for policy purposes, is that the externality is produced by the interaction of the market with infrastructure maintained by a third party. The market is the source of the incentive and the infrastructure is the vulnerable system but the costs are borne elsewhere. The market operator's private optimization does not require internalizing the externality, because the externality falls on parties outside the market relationship.

3 Generalization and Scope

The Météo-France case is specific, but the mechanism it illustrates is general. To understand where the oracle externality poses practical risk rather than just structural possibility, it is useful to characterize the features that made the weather station attack so effective. We posit three factors determine practical severity, namely the attack cost relative to contract payout, the detection latency of the manipulation, and the reversibility of the oracle reading once produced. The weather station case scores maximally on all three with negligible cost, days-long detection latency, and irreversible resolution, which is what made a \$14,000 payout on a trivial stake possible.

Environmental and climate monitoring infrastructure broadly shares these properties. Air quality monitors, hydrological sensors, and remote sensing ground truth sites are physically accessible, publicly documented, and designed for passive observation under non-adversarial assumptions. If prediction markets expand into climate-adjacent contracts, these systems present attack surfaces structurally identical to the Météo-France case. At the other end of the spectrum, markets on macro outcomes (total US GDP, global temperature anomalies) resist individual manipulation because the oracle is an aggregation too distributed for any individual actor to influence. These markets may produce negligible oracle externalities regardless of their volume.

Table 1. Severity of the oracle externality across domains, scored on attack cost relative to payout, detection latency, and reversibility. The agent-mediated case occupies the same corner as the weather sensor, with attack cost driven lower still.

Domain	Attack cost	Detection	Reversibility
Weather sensor (single source)	Very low	Slow	Irreversible
Public-health data	Low	Very slow	Irreversible
Sports outcomes	Moderate	Fast	Irreversible
Electoral infrastructure	Moderate	Fast	Reversible
Macro aggregates (GDP, global temp.)	Prohibitive	—	—
Agentic / public-API oracle	Minimal	Slow	Irreversible

Between these poles sit domains where the three factors interact in more complex ways. Electoral infrastructure has evolved substantial defensive capacity [1, 2] and detection latency is short due to intense concurrent scrutiny, but attack cost may not be high relative to multi-million-dollar contract payouts [14]. Sports outcomes represent a case where the externality is real but partially internalized as defensive infrastructure has developed in response to gambling-driven match-fixing threats [10, 13], though it is funded by sports organizations rather than by the betting operators whose volume imposes costs on it. Public health data presents a distinctive concern because detection latency could be very long and defensive capacity is minimal, so the vulnerability profile may more closely resemble the weather station case than the election case. The commodity market analogue is instructive. The LME aluminum warehouse episode [4] exhibited precisely the same structure of modest attack costs, long detection latency, and irreversible effects on physical delivery.

The common structural feature across these domains remains that the infrastructure predates the prediction markets, was designed for purposes other than serving as market oracles, and is maintained by entities that receive no compensation from the markets that use their data. But the practical severity of the oracle externality varies enormously depending on attack cost, detection latency, and oracle reversibility, and Table 1 summarizes where these domains fall. The weather sensor case is not the most consequential instance but it is timely and clean.

The mechanism does not, however, require the manipulated infrastructure to be a physical instrument. Its three conditions, namely an automated process whose payoff depends on a specific value of externally produced data, a third party producing that data under a non-adversarial threat model, and the absence of any relationship through which the cost is internalized, are satisfied whenever a software agent acts on data from a public endpoint it does not control. As autonomous agents that read third-party APIs and open data feeds proliferate, the population of parties able to act on a manipulation incentive grows, and the attack cost factor is driven toward its floor, since where the Paris attacker had to drive to an airport, an agent can exercise an endpoint at negligible marginal cost. The case to distinguish is load-driven degradation, where agents merely issue automated traffic against endpoints provisioned for human-scale use, which is a commons problem rather than our mechanism because the agent wants the data and not any particular value of it. The oracle externality proper is the specific-value variant, where a pricing agent profits when a competitor’s feed reports a particular number, or an arbitrage agent’s payoff is keyed to a public index, so that the agent’s operator occupies the structural position of the Polymarket bettor while the hardening cost lands on an endpoint operator with no contractual tie to it.

This generalization is adjacent to a growing literature on the security of LLM-based agents, but points in the opposite direction, as that work studies the agent as victim through indirect prompt injection [8, 12], memory manipulation against agents transacting in financial protocols [16], and information-flow defenses against compromised agents [6], with the costs of compromise falling inside the attacker-agent-user triangle. The oracle externality is the dual case, in which the agent’s operator is the beneficiary and the cost falls outside the relationship on the infrastructure operator, and to our knowledge this framing has not been articulated in either the prediction-market or the agent-security literature. A recurring observation in that work is that conventional monitoring fails on agents because they act independently and at volume with no user present to anchor attribution, which attacks the detection-latency factor directly and pushes an agentic oracle attacker toward the worst case of Table 1. The concern applies specifically to markets where individual or small-group action can affect the oracle reading in ways that profitably shift contract resolution.

Current prediction market regulation addresses harms within the market relationship such as insider trading, manipulation between participants, consumer protection. The oracle externality falls outside this scope because the harmed parties are not party to the market. Internalization would require routing costs from market operators to infrastructure operators, whether through direct compensation, Pigouvian taxation with revenues directed to infrastructure maintenance [17], or outright prohibition of markets on manipulable oracles, and Coasean bargaining is unlikely to resolve the externality because infrastructure operators are typically public sector entities without commercial negotiation mandates [5].

A more tractable direction might be oracle design requirements at the platform level, but it is important to be precise about which requirements reach the externality. The major platforms already operate dispute mechanisms, from Polymarket’s UMA optimistic oracle, in which a proposed outcome stands unless challenged within a short window before escalating to a token-holder vote [18], to Kalshi’s regulated exchange and Hyperliquid’s pre-specified source with no dispute layer at all. None reached the Météo-France attack, because a dispute layer adjudicates whether a proposed outcome correctly reports the oracle’s reading, not whether the reading itself was corrupted at the source. A challenger to the Paris market would have lost, since the sensor actually did read above the threshold; the challenge window protects against misreporting the data, which is a harm inside the market relationship, and is silent on manipulation of the data, which is the externality. That these mechanisms are independently strained, with a substantial share of disputed-market votes coming from wallets holding positions in the contract [20], suggests the dispute layer is the wrong place to look.

The requirements that do reach the externality act before resolution, at the data layer. Resolution on multiple independent sources raises manipulation to the cost of a coordinated attack, and resolution on a spatial or temporal aggregate, such as a daily mean across stations rather than a single instantaneous reading, removes the point-in-time target the Paris attack exploited. These have financial market analogs in reference-rate standards and circuit breakers, require no cross-jurisdictional cooperation, and transfer directly to the agentic case, where multi-source resolution raises an automated attacker’s cost just as it does a physical one. The framing opens corresponding empirical and formal directions, namely estimating externality magnitudes in specific markets and modeling equilibrium behavior under different source-design regimes. This paper’s limitation is that it articulates the mechanism and the lever without resolving either, and these are the natural next steps.

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